

PLSC 504 – Fall 2026

Panel/TSCS Data: Unit Effects (+ Dynamics)

October 5, 2026

Terminology:

- “Unit” / “Units” / “Units of observation” / “Panels” = Things we observe repeatedly
- “Observations” = Each (one) measurement of a unit
- “Time points” = When each observation on a unit is made
- $i \in \{1 \dots N\}$ indexes units
- $t \in \{1 \dots T\}$ or $\{1 \dots T_i\}$ indexes observations / time points
- If $T_i = T \forall i$ then we have “balanced” panels / units
- NT = Total number of observations (if balanced)

More terms:

- $N \gg T \rightarrow$ “panel” data
 - NES panel studies ($N \approx 2000, T = 3$)
 - Panel Study of Income Dynamics ($N = \text{large}, T \approx 12$)
- $T \gg N$ or $T \approx N \rightarrow$ “time-series cross-sectional” (“TSCS”) data
- $N = 1 \rightarrow$ “time series” data

Two-Way Variation

Two “dimensions” of variation:

- Cross-sectional variation: how each unit is – on average – different from other units – a/k/a **between-unit** variation
- Temporal variation: how each measurement / time point is – on average – different from other time points on average – a/k/a/ **within-unit** variation

In panel data, a random variable may:

- Have only between-unit variation (i.e., lack *temporal variation*)
- Have only within-unit variation (i.e., lack *cross-sectional variation*)
- Have *both* within- and between-unit variation

For Y_{it} , a variable that varies over both units and time:

- $\bar{Y}_i = \frac{1}{T_i} \sum_{t=1}^T Y_{it}$ is the over-time mean of Y for unit i ,
- $\bar{Y}_t = \frac{1}{N_t} \sum_{i=1}^N Y_{it}$ is the across-unit mean of Y at time t , and
- $\bar{Y} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T Y_{it}$ is the grand mean of Y .

Within- and Between-Unit Variation

The *within-unit* mean of Y is:

$$\bar{Y}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} Y_{it}$$

That means that we can write:

$$Y_{it} = \bar{Y}_i + (Y_{it} - \bar{Y}_i).$$

That is, the *total* variation in Y_{it} can be decomposed into:

- The *between-unit* variation in the $\bar{Y}_i$ s, and
- The *within-unit* variation around $\bar{Y}_i$ (that is, $Y_{it} - \bar{Y}_i$).

Note that (while unusual) one could do the same decomposition vis-à-vis time:

$$Y_{it} = \bar{Y}_t + (Y_{it} - \bar{Y}_t).$$

In a similar fashion, we can also calculate the within- and between-unit variability (e.g., the standard deviations) of the constituent variables $\bar{Y}_i$ and $(Y_{it} - \bar{Y}_i)$...

Our “Running Example” Data: WDI, 1960-2025

The [World Development Indicators](#)

- Cross-national country-level panel data
- $N = 215$ countries, $T = 66$ years (1960-2025) + missingness
- Variable types:
 - Geography: land area, arable land
 - Population indicators
 - Demographics: Birth rates, life expectancy, etc.
 - Economics: GDP, inflation, trade, FDI, etc.
 - Governments: expenditures, policies, etc.
- Full descriptions are listed in the Github repo [here](#).

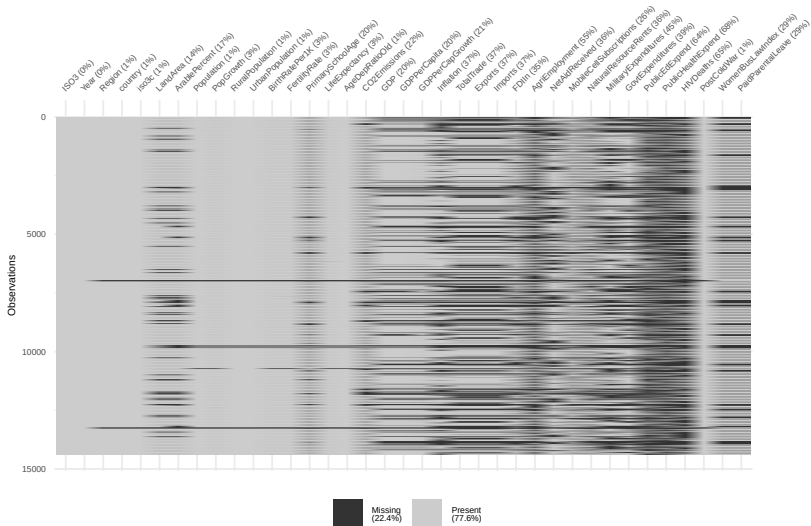
Summary Statistics: WDI Data

```
> describe(wdi,fast=TRUE,ranges=FALSE,check=TRUE)
```

	vars	n	mean	sd	skew	kurtosis	se
ISO3	1	14364	NaN	NA	NA	NA	NA
Year	2	14364	1992.54	1.9e+01	0.00	-1.20	0.16
Region	3	14256	NaN	NA	NA	NA	NA
country	4	14256	NaN	NA	NA	NA	NA
iso3c	5	14256	NaN	NA	NA	NA	NA
LandArea	6	12288	596587.32	1.6e+06	5.46	34.81	14839.81
ArablePercent	7	11871	13.40	1.4e+01	1.47	1.96	0.13
Population	8	14160	25727055.67	1.1e+08	9.72	105.50	897577.36
PopGrowth	9	13943	1.73	1.8e+00	0.73	21.16	0.02
RuralPopulation	10	14190	47.62	2.6e+01	-0.08	-0.99	0.22
UrbanPopulation	11	14190	52.38	2.6e+01	0.08	-0.99	0.22
BirthRatePer1K	12	13945	27.51	1.3e+01	0.25	-1.23	0.11
FertilityRate	13	13943	3.80	2.0e+00	0.45	-1.17	0.02
PrimarySchoolAge	14	11539	6.13	6.1e-01	-0.03	0.13	0.01
LifeExpectancy	15	13941	65.31	1.1e+01	-0.77	0.19	0.10
AgeDepRatioOld	16	14160	10.93	7.4e+00	1.82	5.06	0.06
CO2Emissions	17	11165	5.04	1.1e+01	8.69	102.26	0.11
GDP	18	11500	249018130410.03	1.2e+12	11.60	162.53	11051209271.31
GDPPerCapita	19	11500	12521.98	2.0e+04	3.14	14.85	182.11
GDPPerCapGrowth	20	11377	1.92	6.5e+00	1.69	41.30	0.06
Inflation	21	9113	22.53	3.2e+02	54.78	3719.40	3.32
TotalTrade	22	9011	78.35	5.4e+01	2.97	17.60	0.57
Exports	23	9011	36.46	2.9e+01	2.95	16.10	0.30
Imports	24	9020	41.89	2.8e+01	2.48	13.30	0.29
FDIIn	25	9288	5.43	4.4e+01	15.49	566.30	0.46
AgriEmployment	26	6496	27.25	2.3e+01	0.68	-0.64	0.29
NetAidReceived	27	9218	528849286.43	1.1e+09	11.21	269.53	11599881.51
MobileCellSubscriptions	28	10587	39.15	5.4e+01	1.26	1.61	0.52
NaturalResourceRents	29	9211	6.84	1.1e+01	2.60	8.04	0.12
MilitaryExpenditures	30	7892	2.68	3.2e+00	9.43	238.04	0.04
GovtExpenditures	31	8711	16.45	8.3e+00	3.52	31.41	0.09
PublicEdExpend	32	5125	4.29	2.1e+00	6.10	158.28	0.03
PublicHealthExpend	33	4547	3.31	2.4e+00	1.37	3.43	0.04
HIVDeaths	34	5020	7002.28	2.1e+04	5.74	45.55	294.03
PostColdWar	35	14256	0.55	5.0e-01	-0.18	-1.97	0.00
WomenBusLawIndex	36	10260	59.97	1.9e+01	0.01	-0.59	0.19
PaidParentalLeave	37	10260	0.11	3.1e-01	2.50	4.24	0.00

Missing Data

```
> library(naniar)
```

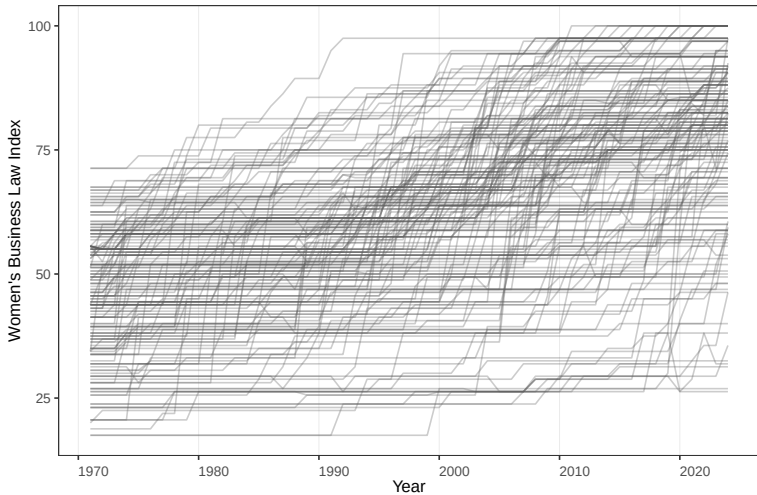


WDI's *Women, Business and the Law Index* (WBLI)

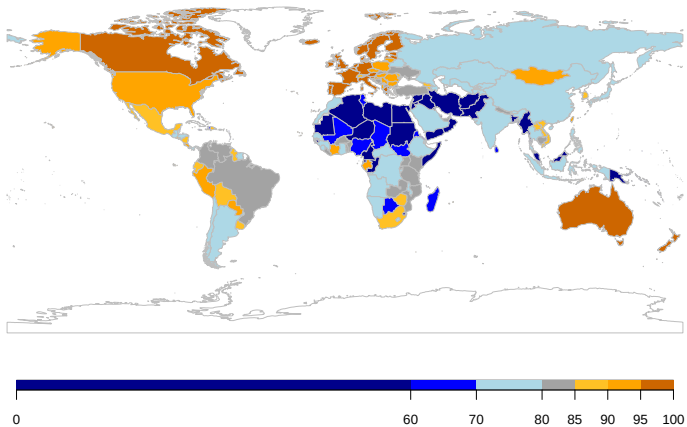
The basis for World Bank reports in [2021](#) (and [2026](#))...

- Version 1.0 examines “the laws and regulations that affect women's economic opportunity in 190 economies” from 1971-2024.
- An index comprising eight indicators “structured around women's interactions with the law as they move through their careers: *Mobility, Workplace, Pay, Marriage, Parenthood, Entrepreneurship, Assets, and Pension.*”
- The WBL Index:
 - Theoretically ranges from 0 - 100
 - In practice: lowest values $\approx$ 18
 - Higher values correspond to higher levels of women's empowerment and greater opportunities and support for women, particularly in business
- “Better performance in the areas measured by the *Women, Business and the Law Index* is associated with a more narrow gender gap in development outcomes, higher female labor force participation, lower vulnerable employment, and greater representation of women in national parliaments.”

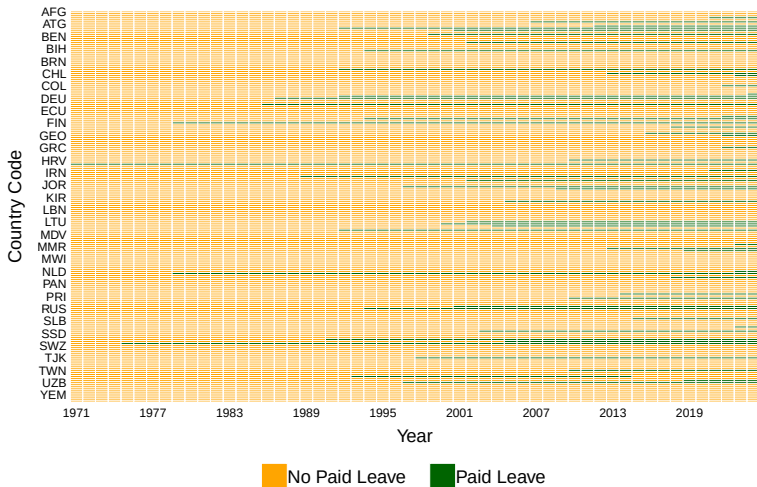
Trend Line Visualization (using panelView)



Map! WBLI (2023)



Categorical Variable Visualization (using panelView)



The model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

makes...

- ...all the usual OLS assumptions, plus
- $\beta_{0i} = \beta_0 \forall i$
- $\beta_{1i} = \beta_1 \forall i$

For:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + u_{it}$$

... it's the same.

Variable Intercepts + Slopes

Relax $\beta_{0i} = \beta_0 \forall i$:

$$Y_{it} = \beta_{0i} + \beta_1 X_{it} + u_{it} \quad (\text{unit-level})$$

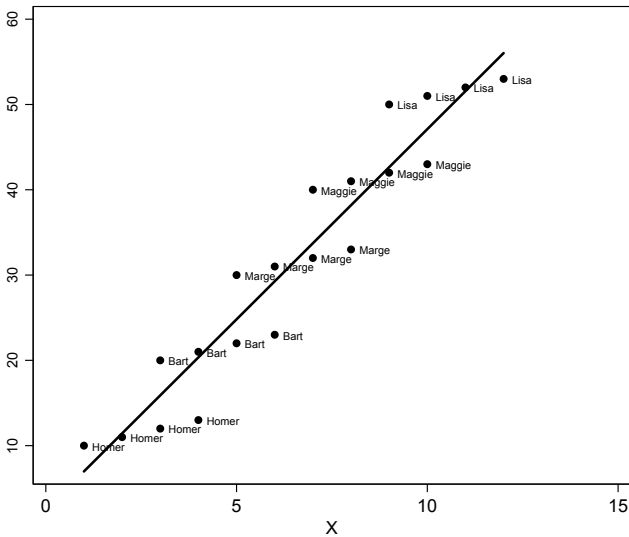
Relax $\beta_{1i} = \beta_1 \forall i$:

$$Y_{it} = \beta_0 + \beta_{1i} X_{it} + u_{it} \quad (\text{unit-level slopes})$$

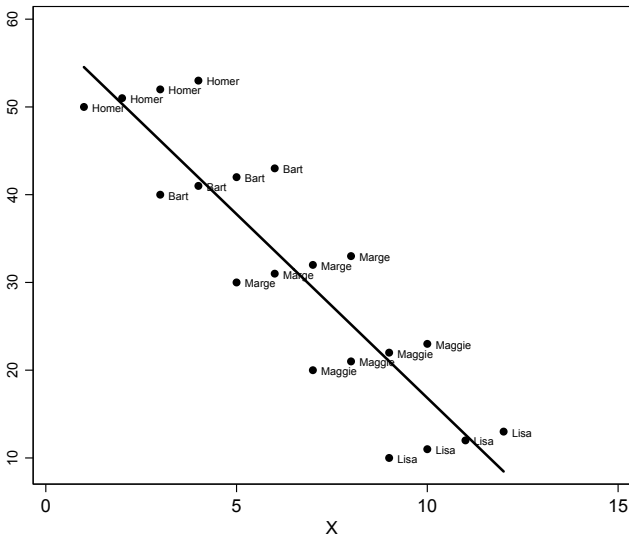
Relax both:

$$Y_{it} = \beta_{0i} + \beta_{1i} X_{it} + u_{it} \quad (\text{unit-level slopes and intercepts})$$

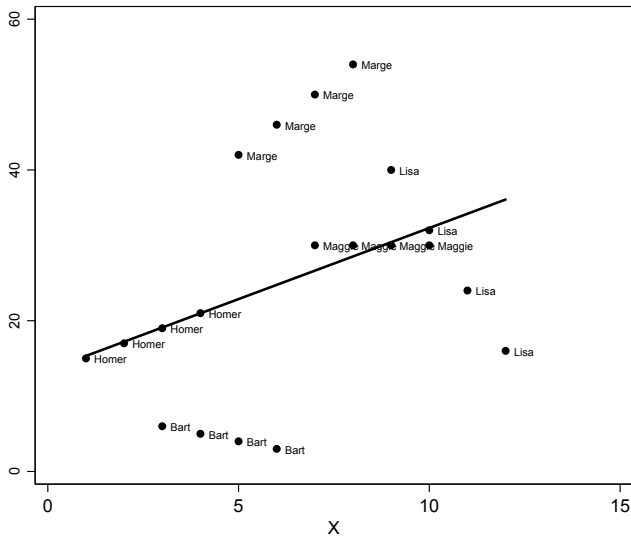
Varying Intercepts



Varying Intercepts



Varying Slopes + Intercepts



Two-Way Variation

Two-way variation:

$$Y_{it} = \mathbf{X}_{it}\beta + \gamma V_i + \delta W_t + u_{it}$$

where V_i are the ℓ predictors which don't vary over time (within a unit), and W_t are the m predictors which don't vary across units (for a given time point).

Note that we can write:

$$\alpha_i = \sum_{\ell} (\gamma V_i)$$

and

$$\eta_t = \sum_m (\delta W_t).$$

So:

$$\begin{aligned} Y_{it} &= \mathbf{X}_{it}\beta + \gamma V_i + \delta W_t + u_{it} \\ &= \mathbf{X}_{it}\beta + \alpha_i + \eta_t + u_{it} \end{aligned}$$

“One-Way” and “Two-Way” Effects

“Two-way” effects:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + \eta_t + u_{it}$$

“One-way” effects:

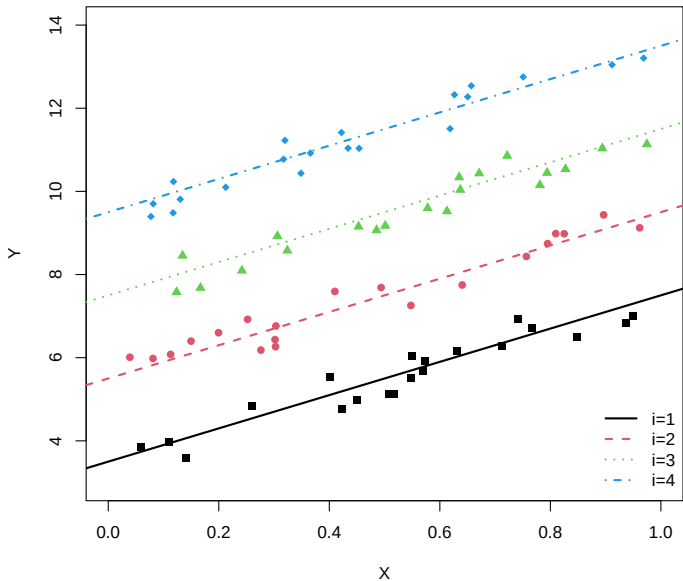
- Assuming $\alpha_i = 0$:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \eta_t + u_{it} \quad (\text{“time effects”})$$

- Assuming $\eta_t = 0$:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + u_{it} \quad (\text{“unit effects”})$$

Intuition: One-Way Unit Effects



(One-Way) “Fixed” Effects

“Brute force” model fits:

$$\begin{aligned}Y_{it} &= \mathbf{X}_{it}\boldsymbol{\beta}_{FE} + \alpha_i + u_{it} \\ &= \mathbf{X}_{it}\boldsymbol{\beta}_{FE} + \alpha_1 I(i = 1)_i + \alpha_2 I(i = 2)_i + \dots + u_{it}\end{aligned}$$

In other words:

- Specify $\mathbf{X}_{it}$,
- Fit a model that includes both $\mathbf{X}_{it}\boldsymbol{\beta}$ and N (or $N - 1$) indicator variables (with parameters α_i), one for each unit i
 - If the model includes an intercept (β_0), then one unit is omitted and becomes the “reference” category
 - If the model omits an intercept, then all N $\hat{\alpha}$ s can be estimated; $\hat{\alpha}_i$ is then the expected value of Y for unit i when all values of $\mathbf{X}_{it}$ are zero

“Fixed” Effects (continued)

Alternatively, decompose:

$$\bar{\mathbf{X}}_i = \frac{\sum_{N_i} \mathbf{X}_{it}}{N_i}$$

and

$$\tilde{\mathbf{X}}_{it} = \mathbf{X}_{it} - \bar{\mathbf{X}}_i.$$

Yields:

$$Y_{it} = \bar{\mathbf{X}}_i \beta_B + \tilde{\mathbf{X}}_{it} \beta_W + \alpha_i + u_{it}$$

But!

$$\text{corr}(\bar{\mathbf{X}}_i \beta_B, \alpha_i) = 1.0$$

“Fixed” Effects = “Within” Effects

This means that:

$$\begin{aligned}Y_{it}^* &= Y_{it} - \bar{Y}_i \\ \mathbf{X}_{it}^* &= \mathbf{X}_{it} - \bar{\mathbf{X}}_i\end{aligned}$$

gives:

$$Y_{it}^* = \mathbf{X}_{it}^* \beta_{FE} + u_{it}.$$

→ **A “fixed effects” model is actually a “within-effects” model.**

- “Fixed effects” models only use the *within-unit* variation in Y and $\mathbf{X}$
- $\hat{\beta}$ s for variables that do not vary within units cannot be estimated

“Fixed” Effects: Test(s)

The one-way FE model implies that $\alpha_i \neq \alpha_j$ for at least some $i \neq j$. We can test this assumption using a standard F -test of the hypothesis:

$$H_0 : \alpha_i = \alpha_j \forall i \neq j$$

versus

$$H_A : \alpha_i \neq \alpha_j \text{ for some } i \neq j$$

Specifically, the test statistic

$$F = \frac{(\text{SSE}_{OLS} - \text{SSE}_{FE}) / N - 1}{\text{SSE}_{FE} / [NT - (N - 1)]}$$

is $\sim F_{N-1, NT-(N-1)}$.

`plm` (panel linear models) is the workhorse R package for fitting linear regression models to panel data. To do so, you must first declare the data as panel data and indicate the variables containing the unit (i) and period (t) indices:

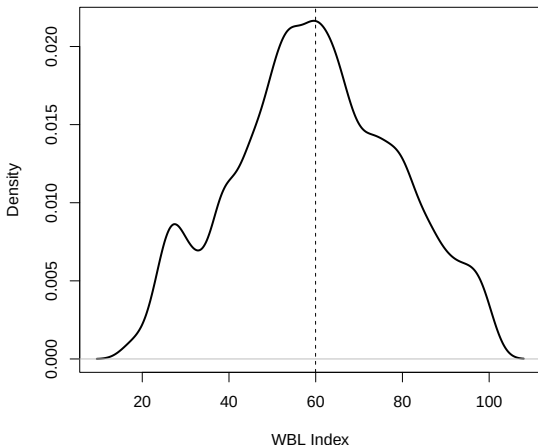
```
> WDI<-pdata.frame(wdi,index=c("ISO3","Year"))
> class(WDI)
[1] "pdata.frame" "data.frame"

> WBLI<-WDI$WomenBusLawIndex
> class(WBLI)
[1] "pseries" "numeric"
```

WBLI: Total Variation

```
> describe(WBLI,na.rm=TRUE) # all variation
```

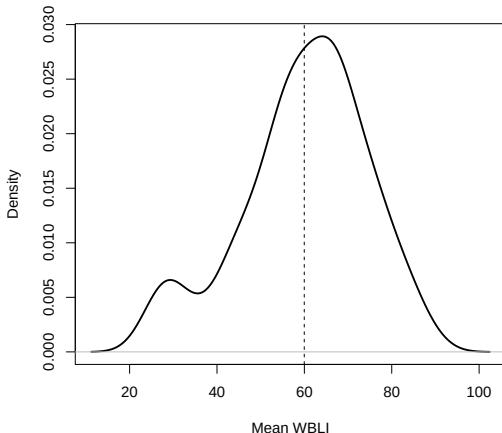
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	10260	60	19	59	60	19	18	100	82	0.01	-0.59	0.19



WBLI: "Between" Variation

```
> describe(plm:::between(WBLI,effect="individual",na.rm=TRUE)) # "between" variation
```

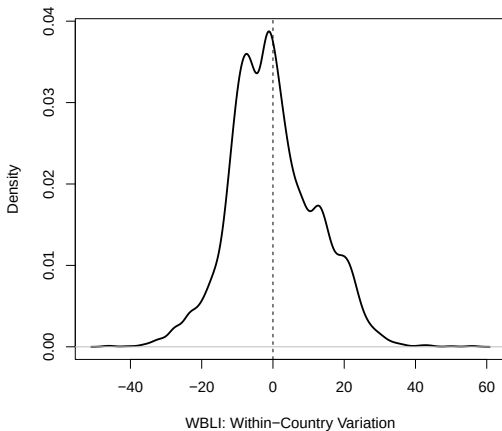
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	190	60	14	61	61	13	24	90	66	-0.51	-0.11	1



WBLI: “Within” Variation

```
> describe(Within(WBLI,na.rm=TRUE)) # "within" variation
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	10260	0	12	-1.2	-0.33	11	-46	56	102	0.26	0.32	0.12



A Regression Model

Regression model:

$$\text{WBLI}_{it} = \beta_0 + \beta_1 \text{Population Growth}_{it} + \beta_2 \text{Urban Population}_{it} + \beta_3 \text{Fertility Rate}_{it} + \beta_4 \ln(\text{GDP Per Capita})_{it} + \beta_5 \text{Natural Resource Rents}_{it} + \beta_6 \text{Post Cold War}_t + u_{it}$$

$$\begin{aligned} \text{Expectations: } \beta_1 &< 0 & \beta_3 &< 0 & \beta_5 &< 0 \\ \beta_2 &> 0 & \beta_4 &> 0 & \beta_6 &> 0 \end{aligned}$$

Descriptive Statistics:

	vars	n	mean	sd	median	min	max	range	skew	kurtosis	se
WomenBusLawIndex	1	8252	60.0	18.85	60.6	17.50	100.0	82.5	-0.02	-0.66	0.21
PopGrowth	2	8252	1.7	1.67	1.6	-27.47	21.7	49.2	-0.09	30.12	0.02
UrbanPopulation	3	8252	52.5	23.88	53.0	3.01	100.0	97.0	0.03	-1.05	0.26
FertilityRate	4	8252	3.6	1.89	3.1	0.77	8.6	7.8	0.53	-1.01	0.02
NaturalResourceRents	5	8252	7.3	11.18	2.5	0.00	88.6	88.6	2.49	7.40	0.12
PostColdWar	6	8252	0.7	0.46	1.0	0.00	1.0	1.0	-0.87	-1.24	0.01
lnGDPPerCap	7	8252	8.3	1.46	8.2	4.93	11.7	6.7	0.11	-0.89	0.02

Where is the Variation in our Data?

Variable	Dimension	Mean	SD	Min	Max	Observations
Women Business Law Index	overall	60.033	18.849	17.5	100	N = 8252
	between		15.236	23.173	89.361	n = 188
	within		11.177	15.182	118.692	T = 43.894
Population Growth	overall	1.666	1.673	-27.471	21.7	N = 8252
	between		1.27	-1.077	6.328	n = 188
	within		1.143	-29.138	17.381	T = 43.894
Urban Population	overall	52.478	23.876	3.01	100	N = 8252
	between		22.723	8.472	100	n = 188
	within		7.073	15.191	85.194	T = 43.894
Fertility Rate	overall	3.59	1.887	0.772	8.606	N = 8252
	between		1.662	1.292	7.588	n = 188
	within		0.916	0.874	6.98	T = 43.894
Natural Resource Rents	overall	7.284	11.184	0	88.592	N = 8252
	between		10.06	0	43.634	n = 188
	within		5.279	-23.94	61.391	T = 43.894
Post-Cold War	overall	0.7	0.458	0	1	N = 8252
	between		0.154	0.345	1	n = 188
	within		0.439	-0.27	1.355	T = 43.894
ln(GDP Per Capita)	overall	8.34	1.457	4.926	11.671	N = 8252
	between		1.394	5.682	11.199	n = 188
	within		0.357	6.528	10.257	T = 43.894

Regression: Pooled OLS

```
> OLS<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+  
+          NaturalResourceRents+PostColdWar,data=WDI,model="pooling")
```

```
>  
> summary(OLS)  
Pooling Model
```

Call:

```
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +  
      FertilityRate + log(GDPPerCapita) + NaturalResourceRents +  
      PostColdWar, data = WDI, model = "pooling")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-78.29	-8.62	1.01	9.27	45.19

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	51.9633	1.7857	29.10	< 2e-16 ***
PopGrowth	-1.7837	0.1165	-15.31	< 2e-16 ***
UrbanPopulation	-0.0749	0.0104	-7.22	5.7e-13 ***
FertilityRate	-2.9900	0.1549	-19.31	< 2e-16 ***
log(GDPPerCapita)	2.5588	0.1878	13.62	< 2e-16 ***
NaturalResourceRents	-0.3506	0.0150	-23.32	< 2e-16 ***
PostColdWar	9.8824	0.3659	27.01	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 2930000

Residual Sum of Squares: 1490000

R-Squared: 0.493

Adj. R-Squared: 0.493

F-statistic: 1337.18 on 6 and 8245 DF, p-value: <2e-16

"Fixed" (Within) Effects

```
> FE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+
+         NaturalResourceRents+PostColdWar,data=WDI,effect="individual",model="within")
>
> summary(FE)
Oneway (individual) effect Within Model
```

```
Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
     PostColdWar, data = WDI, effect = "individual", model = "within")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Residuals:

	Min.	1st Qu.	Median	3rd Qu.	Max.
	-33.766	-5.090	-0.628	4.982	55.202

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
PopGrowth	-0.1303	0.0806	-1.62	0.11
UrbanPopulation	0.3370	0.0183	18.41	< 2e-16 ***
FertilityRate	-1.5477	0.1549	-9.99	< 2e-16 ***
log(GDPPerCapita)	8.5685	0.2895	29.60	< 2e-16 ***
NaturalResourceRents	0.0898	0.0166	5.41	0.000000066 ***
PostColdWar	6.7013	0.2860	23.43	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 1030000
Residual Sum of Squares: 495000
R-Squared: 0.52
Adj. R-Squared: 0.508
F-statistic: 1454.48 on 6 and 8058 DF, p-value: <2e-16

Table: Models of WBLI

	OLS	FE
Population Growth	-1.795*** (0.117)	-0.142* (0.080)
Urban Population	-0.074*** (0.010)	0.336*** (0.018)
Fertility Rate	-2.994*** (0.155)	-1.543*** (0.155)
ln(GDP Per Capita)	2.537*** (0.188)	8.588*** (0.289)
Natural Resource Rents	-0.350*** (0.015)	0.090*** (0.017)
Post-Cold War	9.869*** (0.366)	6.701*** (0.286)
Constant	52.130*** (1.785)	
Observations	8,252	8,252
R ²	0.493	0.520
Adjusted R ²	0.493	0.509

* p<0.1; ** p<0.05; *** p<0.01

As we noted previously, the specification:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + u_{it}$$

...suggests that we can use an F -test to examine the hypothesis:

$$H_0 : \alpha_i = \alpha \forall i$$

(and a similar test for $\eta_t = 0$ in the time-centered case).

Arguably better are Lagrange multiplier-based tests:

- Breusch-Pagan (1980)
- King and Wu (1997)
- See (e.g.) Croissant and Millo (2018, §4.1) for details

FE (by Unit) Model Tests

```
> pFtest(FE,OLS)
```

F test for individual effects

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
F = 86, df1 = 187, df2 = 8058, p-value <2e-16  
alternative hypothesis: significant effects
```

```
> plmtest(FE,effect=c("individual"),type=c("bp"))
```

Lagrange Multiplier Test - (Breusch-Pagan)

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
chisq = 55552, df = 1, p-value <2e-16  
alternative hypothesis: significant effects
```

```
> plmtest(FE,effect=c("individual"),type=c("kw"))
```

Lagrange Multiplier Test - (King and Wu)

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
normal = 236, p-value <2e-16  
alternative hypothesis: significant effects
```

Fixed Effects: Interpretation

FE models are *subject-specific* models, that rely entirely on *within-unit* variability to estimate variable effects.

- This means that:

$$\hat{\beta}_k = \frac{\partial E(Y|\hat{\alpha})}{\partial X_k}$$

- That is, $\hat{\beta}_k$ is the expected change in $E(Y)$ associated with a one-unit increase in observation i 's value of X_k
- Key: *within-unit* changes in $\mathbf{X}$ are associated with *within-unit* expected changes in Y .
- In a linear model, the value of $\hat{\alpha}$ doesn't affect the value of that partial derivative...

Fixed Effects: Interpretation

Mummolo and Peterson (2018) note that:

“...because the within-unit variation is always smaller (or at least, no larger) than the overall variation in the independent variable, researchers should use within-unit variation to motivate counterfactuals when discussing the substantive impact of a treatment” (2018, 829).

Significance:

- Predictors **X** in FE models typically have both cross-sectional and temporal variation
- FE models only consider *within-unit* variation in **X** and *Y*
- As a result, the degree of actual/observed (within-unit) variation in predictors is almost always less – and sometimes significantly less – than if cross-sectional variation were also considered

Interpretation Example: Urban Population

Q: How much is a “one standard deviation change in Urban Population”?

Recall its variation:

Variable	Dimension	Mean	SD	Min	Max	Observations
Urban Population	overall	52.478	23.876	3.01	100	N = 8252
	between		22.723	8.472	100	n = 188
	within		7.073	15.191	85.194	T = 43.894

“While the overall variation in the independent variable may be large, the within-unit variation used to estimate β may be much smaller” (M & P 2018, p. 830).

Pros and Cons of “Fixed” Effects

Pros:

- Mitigates (Some) Specification Bias
- Simple + Intuitive
- Widely Used/Understood

Cons (see e.g. [Collischon and Eberl 2020](#)):

- Can't Estimate β_B
- Slowly-Changing $\mathbf{X}$ s
- (In)Efficiency / Inconsistency (Incidental Parameters)
- Cannot Control for (e.g.) Time-Varying Heterogeneity
- Sensitivity to Measurement Error

From the equation above:

$$Y_{it} = \bar{\mathbf{X}}_i \beta_B + \tilde{\mathbf{X}}_{it} \beta_W + \alpha_i + u_{it}$$

...we can derive a “Between Effects” model:

$$\bar{Y}_i = \bar{\mathbf{X}}_i \beta_B + u_{it}$$

This model:

- is essentially cross-sectional,
- is based on N observations,
- considers *only* between-unit (average) differences
- Interpretation:

$\hat{\beta}_B$ is the expected difference in Y between two units whose values on $\bar{X}$ differ by a value of 1.0.

"Between" Effects

```
> BE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+
+         log(GDPPerCapita)+NaturalResourceRents+PostColdWar,data=WDI,
+         effect="individual",model="between")
>
> summary(BE)
Oneway (individual) effect Between Model
```

```
Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
    FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
    PostColdWar, data = WDI, effect = "individual", model = "between")
```

```
Unbalanced Panel: n = 188, T = 8-51, N = 8252
Observations used in estimation: 188
```

```
Residuals:
    Min. 1st Qu.  Median 3rd Qu.    Max.
-27.906  -6.913   0.557   8.149  21.382
```

```
Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
(Intercept)    57.5110   12.5965    4.57 0.0000092 ***
PopGrowth      -5.0016    1.1144   -4.49 0.0000128 ***
UrbanPopulation -0.0668    0.0546   -1.22  0.2224
FertilityRate   -1.5507    1.1546   -1.34  0.1809
log(GDPPerCapita) 2.1140    1.1114    1.90  0.0587 .
NaturalResourceRents -0.2991  0.0905   -3.30  0.0012 **
PostColdWar      6.5059    5.3990    1.21  0.2298
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Total Sum of Squares:    43400
Residual Sum of Squares: 18300
R-Squared:      0.579
Adj. R-Squared: 0.565
F-statistic: 41.505 on 6 and 181 DF, p-value: <2e-16
```

A Nicer Table (Again)

Table: Models of WBLI

	OLS	FE	BE
Population Growth	-1.795*** (0.117)	-0.142* (0.080)	-5.002*** (1.114)
Urban Population	-0.074*** (0.010)	0.336*** (0.018)	-0.067 (0.055)
Fertility Rate	-2.994*** (0.155)	-1.543*** (0.155)	-1.551 (1.155)
ln(GDP Per Capita)	2.537*** (0.188)	8.588*** (0.289)	2.114* (1.111)
Natural Resource Rents	-0.350*** (0.015)	0.090*** (0.017)	-0.299*** (0.091)
Post-Cold War	9.869*** (0.366)	6.701*** (0.286)	6.506 (5.399)
Constant	52.130*** (1.785)		57.510*** (12.600)
Observations	8,252	8,252	188
R ²	0.493	0.520	0.579
Adjusted R ²	0.493	0.509	0.565

*p<0.1; **p<0.05; ***p<0.01

“Random” Effects

Model:

$$Y_{it} = \mathbf{X}_{it}\beta + u_{it}$$

with:

$$u_{it} = \alpha_i + \lambda_t + \eta_{it}$$

and

$$\begin{aligned} E(\alpha_i) &= E(\lambda_t) = E(\eta_{it}) &= 0, \\ E(\alpha_i \lambda_t) &= E(\alpha_i \eta_{it}) = E(\lambda_t \eta_{it}) &= 0, \\ E(\alpha_i \alpha_j) &= \sigma_\alpha^2 \text{ if } i = j, \text{ 0 otherwise,} \\ E(\lambda_t \lambda_s) &= \sigma_\lambda^2 \text{ if } t = s, \text{ 0 otherwise,} \\ E(\eta_{it} \eta_{js}) &= \sigma_\eta^2 \text{ if } i = j, \text{ } t = s, \text{ 0 otherwise,} \\ E(\alpha_i \mathbf{X}_{it}) &= E(\lambda_t \mathbf{X}_{it}) = E(\eta_{it} \mathbf{X}_{it}) &= 0. \end{aligned}$$

“Random” Effects

If those assumptions are met, we can consider the “two-way variance components” model where:

$$\begin{aligned}\text{Var}(u_{it}) &= \text{Var}(Y_{it}|\mathbf{X}_{it}) \\ &= \sigma_{\alpha}^2 + \sigma_{\lambda}^2 + \sigma_{\eta}^2\end{aligned}$$

If we assume $\lambda_t = 0$, then we get a model like:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + \eta_{it}$$

with total error variance:

$$\sigma_u^2 = \sigma_{\alpha}^2 + \sigma_{\eta}^2.$$

a.k.a. the “one-way error components model.”

“Random” Effects: Estimation

The model above will violate the standard OLS assumptions of uncorrelated errors, because the (compound) “errors” u_{it} within each unit share a common component α_i . Consider the within- i variance-covariance matrix of the errors $\mathbf{u}$:

$$\begin{aligned} E(\mathbf{u}_i \mathbf{u}_i') \equiv \boldsymbol{\Sigma}_i &= \sigma_\eta^2 \mathbf{I}_T + \sigma_\alpha^2 \mathbf{\bar{1}} \mathbf{\bar{1}}' \\ &= \begin{pmatrix} \sigma_\eta^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\eta^2 + \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\eta^2 + \sigma_\alpha^2 \end{pmatrix} \end{aligned}$$

Assuming conditional independence across units, we then have:

$$\text{Var}(\mathbf{u}) \equiv \boldsymbol{\Omega} = \begin{pmatrix} \boldsymbol{\Sigma}_1 & 0 & \cdots & 0 \\ 0 & \boldsymbol{\Sigma}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{\Sigma}_N \end{pmatrix}$$

“Random” Effects: Estimation

Under our assumptions, we can show that:

$$\Sigma^{-1/2} = \frac{1}{\sigma_\eta} \left[\mathbf{I}_T - \left(\frac{\theta}{T} \mathbf{ii}' \right) \right]$$

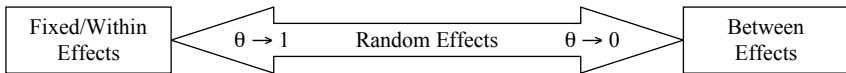
where

$$\theta = 1 - \sqrt{\frac{\sigma_\eta^2}{T\sigma_\alpha^2 + \sigma_\eta^2}}$$

is an unknown quantity to be estimated.

Estimation: Via MLE, or iterated FGLS.

“Random” Effects: An Alternative View



Random Effects

```
> RE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
+         PostColdWar,data=WDI,effect="individual",model="random")
>
> summary(RE)
Oneway (individual) effect Random Effect Model
(Swamy-Arora's transformation)

Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
     PostColdWar, data = WDI, effect = "individual", model = "random")

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Effects:
              var std.dev share
idiosyncratic 61.41   7.84  0.39
individual    95.61   9.78  0.61
theta:
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.727   0.877   0.888   0.881   0.888   0.888

Residuals:
      Min.  1st Qu.  Median    Mean 3rd Qu.    Max.
-33.9275  -5.3225  -0.5789  -0.0308   5.5470  46.1852

Coefficients:
              Estimate Std. Error z-value Pr(>|z|)
(Intercept)   -8.7087    2.4980   -3.49  0.00049 ***
PopGrowth      -0.1913    0.0824   -2.32  0.02024 *
UrbanPopulation  0.2267    0.0174   13.01 < 2e-16 ***
FertilityRate  -1.8838    0.1551  -12.15 < 2e-16 ***
log(GDPPerCapita) 7.0085    0.2744   25.54 < 2e-16 ***
NaturalResourceRents 0.0580    0.0167    3.48  0.00050 ***
PostColdWar     7.8811    0.2820   27.95 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    1070000
Residual Sum of Squares: 534000
R-Squared:              0.501
Adj. R-Squared:         0.501
Chisq: 8107.26 on 6 DF, p-value: <2e-16
```

A Nicer Table (Yet Again)

Table: Models of WBLI

	OLS	FE	BE	RE
Population Growth	-1.795*** (0.117)	-0.142* (0.080)	-5.002*** (1.114)	-0.203** (0.082)
Urban Population	-0.074*** (0.010)	0.336*** (0.018)	-0.067 (0.055)	0.226*** (0.017)
Fertility Rate	-2.994*** (0.155)	-1.543*** (0.155)	-1.551 (1.155)	-1.878*** (0.155)
ln(GDP Per Capita)	2.537*** (0.188)	8.588*** (0.289)	2.114* (1.111)	7.025*** (0.274)
Natural Resource Rents	-0.350*** (0.015)	0.090*** (0.017)	-0.299*** (0.091)	0.058*** (0.017)
Post-Cold War	9.869*** (0.366)	6.701*** (0.286)	6.506 (5.399)	7.881*** (0.282)
Constant	52.130*** (1.785)		57.510*** (12.600)	-8.840*** (2.497)
Observations	8,252	8,252	188	8,252
R ²	0.493	0.520	0.579	0.501
Adjusted R ²	0.493	0.509	0.565	0.501

*p<0.1; **p<0.05; ***p<0.01

“Random” Effects: Testing

Intuition:

- RE models require that $\text{Cov}(X_{it}, \alpha_i) = 0$.
- FE models do not.

This means that:

Model	<u>Reality</u>	
	$\text{Cov}(X_{it}, \alpha_i) = 0$	$\text{Cov}(X_{it}, \alpha_i) \neq 0$
Fixed Effects	Consistent, Inefficient	Consistent, Efficient
Random Effects	Consistent, Efficient	Inconsistent

The Hausman Test

Hausman test (FE vs. RE):

$$\widehat{W} = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})'(\hat{\mathbf{V}}_{\text{FE}} - \hat{\mathbf{V}}_{\text{RE}})^{-1}(\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})$$

$$W \sim \chi_k^2$$

Null: The RE model is consistent ($\text{Cov}(X_{it}, \alpha_i) = 0$).

Issues / Concerns:

- Asymptotic...
- No guarantee $(\hat{\mathbf{V}}_{\text{FE}} - \hat{\mathbf{V}}_{\text{RE}})^{-1}$ is positive definite
- This is a general specification test...

Hausman Test Results

Hausman test (FE vs. RE):

```
> phtest(FE, RE) # ugh...
```

Hausman Test

```
data:  WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...  
chisq = 73, df = 6, p-value = 1e-13
```

```
alternative hypothesis: one model is inconsistent
```

Things to think about:

- Are we in asymptopia?
- Do we *believe* our model specification?...

Practical “Fixed” vs. “Random” Effects

Factors to consider:

- “Panel” vs. “TSCS” Data
- Nature of the Data-Generating Processes
- Importance of Non-Time-Varying Covariate Effects
- Dimension/Location of the Variance in the Data (N vs. T)

Hybrid Models: Combining Within and Between Effects

Recall that we can decompose the variance of our regression model as:

$$Y_{it} = \bar{\mathbf{X}}_i \beta_B + (\mathbf{X}_{it} - \bar{\mathbf{X}}_i) \beta_W + u_{it} \quad (1)$$

This raises the possibility of fitting the model in (1) directly...

- Simple to estimate (via OLS or other)
- Relatively easy interpretation
- Easy to test $\hat{\beta}_B = \hat{\beta}_W$

Example: Separate effects for within- and between-country *Natural Resource Rents*

- Theory: Countries with large natural resource endowments will have lower WBLI on average, *but*
- When natural resource rents increase *within* a country, that will be associated with higher expected WBLI...

Combining Within- and Between-Effects

Table: BE + WE Model of WBLI

	WEBE.OLS
Population Growth	-1.657*** (0.114)
Urban Population	-0.051*** (0.010)
Fertility Rate	-2.643*** (0.152)
ln(GDP Per Capita)	2.485*** (0.183)
Within-Country Nat. Resource Rents	0.119*** (0.027)
Between-Country Nat. Resource Rents	-0.528*** (0.017)
Post-Cold War	10.540*** (0.359)
Constant	50.660*** (1.744)
Observations	8,252
R ²	0.517
Adjusted R ²	0.517
Residual Std. Error	13.100 (df = 8244)

* p<0.1; ** p<0.05; *** p<0.01

Two important things to remember:

1. Recall the variation in *Natural Resource Rents*:

Variable	Dimension	Mean	SD	Min	Max	Observations
Natural Resource Rents	overall	7.284	11.184	0	88.592	N = 8252
	between		10.06	0	43.634	n = 188
	within		5.279	-23.94	61.391	T = 43.894

It is important to keep this in mind when discussing the relative sizes of effects:

- A one-s.d. increase in `NaturalResourceRents` within a country yields an expected change in Y of $5.3 \times 0.12 = 0.64$
- A one-s.d. difference in `NaturalResourceRents` between two countries yields an expected change in Y of $10.1 \times -0.53 = -5.4$

Interpretation (continued)

2. We can formally test whether $\hat{\beta}_B = \hat{\beta}_W$ for the *Natural Resource Rents* variable:

```
> library(car)
> linearHypothesis(WEBE.OLS,c("NRR.Within=NRR.Between"))
```

Linear hypothesis test:

NRR.Within - NRR.Between = 0

Model 1: restricted model

Model 2: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +
log(GDPPERcapita) + NRR.Within + NRR.Between + PostColdWar

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	8245	1485718				
2	8244	1415406	1	70312	410	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Extension: The “Mundlak Device”

Mundlak (1978)¹ notes that in:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + u_{it}$$

the only portion of α_i that can be correlated with u_{it} must also be correlated with *only* the means of $\mathbf{X}$ for each individual. Thus, a model that includes / “controls for” the unit-level means of $\mathbf{X}_{it}$:

$$Y_{it} = \mathbf{X}_{it}\beta + \bar{\mathbf{X}}_i\gamma + u_{it}$$

...will ensure that the remaining variation in $\mathbf{X}_{it}$ is uncorrelated with u_{it} .

Moreover, a test for $\hat{\gamma} = 0$ is similar (and, as it turns out, superior) to the Hausman test for FE vs. RE.

¹Mundlak, Yair. 1978. “On the Pooling of Time Series and Cross Section Data.” *Econometrica* 46:69-85.

“Mundlak Device” (continued)

```
# Create unit-level means of all the (time-varying) predictors:
```

```
smol$PGBetween<-plm::Between(smol$PopGrowth,effect="individual")
smol$UPBetween<-plm::Between(smol$UrbanPopulation,effect="individual")
smol$FRBetween<-plm::Between(smol$FertilityRate,effect="individual")
smol$GDPBetween<-plm::Between(smol$lnGDPPerCap,effect="individual")
smol$NRRBetween<-plm::Between(smol$NaturalResourceRents,effect="individual")
smol$PCWBetween<-plm::Between(smol$PostColdWar,effect="individual")
```

```
# Regress Y on both using plain-old OLS:
```

```
MD<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+
  NaturalResourceRents+PostColdWar+PGBetween+UPBetween+FRBetween+GDPBetween+
  NRRBetween+PCWBetween,data=smol,effect="individual",model="pooling")
```

FE and Mundlak Models

Table: FE and Mundlak Device

	FE2	MD
Population Growth	-0.142* (0.080)	-0.142 (0.127)
Urban Population	0.336*** (0.018)	0.336*** (0.029)
Fertility Rate	-1.543*** (0.155)	-1.543*** (0.245)
ln(GDP Per Capita)	8.588*** (0.289)	8.588*** (0.458)
Natural Resource Rents	0.090*** (0.017)	0.090*** (0.026)
Post-Cold War	6.701*** (0.286)	6.701*** (0.452)
Between-Country Population Growth		-5.334*** (0.246)
Between-Country Urban Population		-0.381*** (0.031)
Between-Country Fertility Rate		0.868*** (0.326)
Between-Country ln(GDP Per Capita)		-5.913*** (0.503)
Between-Country Nat. Resource Rents		-0.464*** (0.032)
Between-Country Post-Cold War		-0.572 (1.235)
Constant		50.020*** (2.379)
Observations	8,252	8,252
R ²	0.520	0.568
Adjusted R ²	0.509	0.568
F Statistic	1,456.000*** (df = 6; 8058)	904.200*** (df = 12; 8239)

* p<0.1; ** p<0.05; *** p<0.01

"Mundlak Device": Testing

```
> # Testing:
```

```
> linearHypothesis(MD,c("PGBetween = 0",  
+                        "UPBetween = 0",  
+                        "FRBetween = 0",  
+                        "GDPBetween = 0",  
+                        "NRRBetween = 0",  
+                        "PCWBetween = 0"))
```

Linear hypothesis test:

```
PGBetween = 0  
UPBetween = 0  
FRBetween = 0  
GDPBetween = 0  
NRRBetween = 0  
PCWBetween = 0
```

Model 1: restricted model

```
Model 2: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate +  
  lnGDPPerCap + NaturalResourceRents + PostColdWar + PGBetween +  
  UPBetween + FRBetween + GDPBetween + NRRBetween + PCWBetween
```

```
Res.Df Df Chisq Pr(>Chisq)  
1    8245  
2    8239  6 1434    <2e-16 ***  
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Two-Way Unit Effects

Our original decomposition considered “two-way” effects:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + \eta_t + u_{it}$$

This “controls for” both non-time-varying unit-level differences and non-cross-sectionally-varying differences between periods. It also implies that we can use (e.g.) an F -test to examine the hypothesis:

$$H_0 : \alpha_i = \eta_t = 0 \quad \forall i, t$$

...that is, whether adding the (two-way) effects improves the model's fit. We can also consider the partial hypotheses:

$$H_0 : \alpha_i = 0 \quad \forall i$$

and

$$H_0 : \eta_t = 0 \quad \forall t$$

separately.

Two-Way Effects: Good & Bad

The Good:

- Addresses both time-invariant, unit-level heterogeneity and cross-sectionally-invariant, time-point-specific heterogeneity
- May be “fixed” or “random” ...
- Two-way FE is equivalent to differences-in-differences when $X \in \{0, 1\}$ and $T = 2$ (more on that later)

The (Potentially) Bad:

- Ability to control for unobserved heterogeneity depends on (probably invalid) functional form assumptions (Imai and Kim 2020)
- FE *requires* predictors that vary both within and between both units and time points
- RE requires the (additional) assumption that $\text{Cov}(\mathbf{X}_{it}, \eta_t) = \text{Cov}(\alpha_i, \eta_t) = 0$
- Two-way effects models ask a *lot* of your data (effectively fits $N + T + k$ parameters using NT observations)

Example: Two-Way Fixed Effects

```
> TwoWayFE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
  PostColdWar,data=WDI,effect="twoway",model="within")
```

```
> summary(TwoWayFE)
Twoways effects Within Model
```

```
Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
     PostColdWar, data = WDI, effect = "twoway", model = "within")
```

```
Unbalanced Panel: n = 188, T = 8-51, N = 8252
```

```
Residuals:
```

Min.	1st Qu.	Median	3rd Qu.	Max.
-32.05	-3.98	0.23	4.10	45.49

```
Coefficients:
```

	Estimate	Std. Error	t-value	Pr(> t)
PopGrowth	-0.2483	0.0672	-3.70	0.00022 ***
UrbanPopulation	0.0673	0.0160	4.21	2.5e-05 ***
FertilityRate	1.5987	0.1413	11.32	< 2e-16 ***
log(GDPPerCapita)	1.9960	0.2654	7.52	6.0e-14 ***
NaturalResourceRents	0.0529	0.0144	3.67	0.00024 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Total Sum of Squares: 348000
```

```
Residual Sum of Squares: 339000
```

```
R-Squared: 0.0271
```

```
Adj. R-Squared: -0.00234
```

```
F-statistic: 44.5486 on 5 and 8009 DF, p-value: <2e-16
```

Two-Way Effects: Testing

```
> # Two-way effects:
```

```
> pFtest(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+
+       log(GDPPerCapita)+NaturalResourceRents+PostColdWar,data=WDI,
+       effect="twoway",model="within")
```

```
F test for twoways effects
```

```
data: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
F = 115, df1 = 236, df2 = 8009, p-value <2e-16
alternative hypothesis: significant effects
```

```
> plmtest(TwoWayFE,c("twoways"),type=("kw"))
```

```
Lagrange Multiplier Test - two-ways effects (King and Wu)
```

```
data: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 191, p-value <2e-16
alternative hypothesis: significant effects
```

```
> # One-way effects in the two-way model:
```

```
> plmtest(TwoWayFE,c("individual"),type=("kw"))
```

```
Lagrange Multiplier Test - (King and Wu)
```

```
data: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 236, p-value <2e-16
alternative hypothesis: significant effects
```

```
> plmtest(TwoWayFE,c("time"),type=("kw"))
```

```
Lagrange Multiplier Test - time effects (King and Wu)
```

```
data: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...
normal = 92, p-value <2e-16
alternative hypothesis: significant effects
```

Two-Way Random Effects

```
> TwoWayRE<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+log(GDPPerCapita)+NaturalResourceRents+
  PostColdWar,data=WDI,effect="twoway",model="random")
```

```
> summary(TwoWayRE)
Twoways effects Random Effect Model
(Swamy-Arora's transformation)
```

Call:

```
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
  FertilityRate + log(GDPPerCapita) + NaturalResourceRents +
  PostColdWar, data = WDI, effect = "twoway", model = "random")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Effects:

	var	std.dev	share
idiosyncratic	42.327	6.506	0.31
individual	96.039	9.800	0.69
time	0.267	0.516	0.00

theta:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
id	0.7715	0.8981	0.9074	0.9012	0.9074	0.9074
time	0.2342	0.2794	0.3144	0.2993	0.3194	0.3213
total	0.2336	0.2789	0.3139	0.2987	0.3189	0.3209

Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-66.011	-7.177	2.159	-0.239	10.623	35.679

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	-0.14488	0.32965	-0.44	0.66
PopGrowth	-0.21729	0.01027	-21.16	<2e-16 ***
UrbanPopulation	0.18116	0.00224	80.91	<2e-16 ***
FertilityRate	-0.67933	0.02010	-33.80	<2e-16 ***
log(GDPPerCapita)	5.48912	0.03579	153.36	<2e-16 ***
NaturalResourceRents	0.05612	0.00213	26.34	<2e-16 ***
PostColdWar	10.72981	0.04428	242.31	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 2930000

Residual Sum of Squares: 2330000

R-Squared: 0.267

Adj. R-Squared: 0.266

Chisq: 265460 on 6 DF, p-value: <2e-16

Table: Models of WBLI

	OLS	FE	BE	RE	Two-Way FE	Two-Way RE
Population Growth	-1.795*** (0.117)	-0.142* (0.080)	-5.002*** (1.114)	-0.203** (0.082)	-0.248*** (0.067)	-0.217*** (0.010)
Urban Population	-0.074*** (0.010)	0.336*** (0.018)	-0.067 (0.055)	0.226*** (0.017)	0.067*** (0.016)	0.181*** (0.002)
Fertility Rate	-2.994*** (0.155)	-1.543*** (0.155)	-1.551 (1.155)	-1.878*** (0.155)	1.599*** (0.141)	-0.679*** (0.020)
ln(GDP Per Capita)	2.537*** (0.188)	8.588*** (0.289)	2.114* (1.111)	7.025*** (0.274)	1.996*** (0.265)	5.489*** (0.036)
Natural Resource Rents	-0.350*** (0.015)	0.090*** (0.017)	-0.299*** (0.091)	0.058*** (0.017)	0.053*** (0.014)	0.056*** (0.002)
Post-Cold War	9.869*** (0.366)	6.701*** (0.286)	6.506 (5.399)	7.881*** (0.282)		10.730*** (0.044)
Constant	52.130*** (1.785)		57.510*** (12.600)	-8.840*** (2.497)		-0.145 (0.330)
Observations	8,252	8,252	188	8,252	8,252	8,252
R ²	0.493	0.520	0.579	0.501	0.027	0.267
Adjusted R ²	0.493	0.509	0.565	0.501	-0.002	0.266

*p<0.1; **p<0.05; ***p<0.01

Interpretation: `modelsummary` and `marginaleffects`

Two broadly useful packages by [Vincent Arel-Bundock](#):

1. `modelsummary`

- “`modelsummary` creates tables and plots to present descriptive statistics and to summarize statistical models in R.”
- In short, it... summarizes models (and data)

2. `marginaleffects`

- An “R package to compute and plot predictions, slopes, marginal means, and comparisons (contrasts, risk ratios, odds ratios, etc.) for over 70 classes of statistical models in R.”
- Focus is on individual covariates / predictors
- Especially handy for nonlinear models (GLMs, etc.)

Both are highly flexible and customizable, and **both play well with the models fit by `plm`.**

Using modelsummary

The most basic version of the code is:

```
> modelsummary(list(OLS,FE,RE))
```

...from which we get this (in the Viewer window):

	(1)	(2)	(3)
(Intercept)	52.133 (1.785)		-8.840 (2.497)
PopGrowth	-1.795 (0.117)	-0.142 (0.080)	-0.203 (0.082)
UrbanPopulation	-0.074 (0.010)	0.336 (0.018)	0.226 (0.017)
FertilityRate	-2.994 (0.155)	-1.543 (0.155)	-1.878 (0.155)
log(GDPPerCapita)	2.537 (0.188)	8.588 (0.289)	7.025 (0.274)
NaturalResourceRents	-0.350 (0.015)	0.090 (0.017)	0.058 (0.017)
PostColdWar	9.869 (0.366)	6.701 (0.286)	7.881 (0.282)
Num.Obs.	8252	8252	8252
R2	0.493	0.520	0.501
R2 Adj.	0.493	0.509	0.501
AIC	66289.0	57209.3	57833.8
BIC	66345.2	57258.4	57889.9
RMSE	13.42	7.74	8.04

Using modelsummary (continued)

A better-looking table:

```
> modelsummary(models,output="MS-Table-26.tex",title="Models of WDBI",stars=TRUE,fmt=2,  
+   gof_map=c("nobs","r.squared","adj.r.squared"),coef_rename=c("PopGrowth"="Population Growth",  
+   "UrbanPopulation"="Urban Population","FertilityRate"="Fertility Rate","log(GDPPerCapita)"="ln(GDP Per Capita)",  
+   "NaturalResourceRents"="Natural Resource Rents","PostColdWar"="Post-Cold War"))
```

Table 7: Models of WDBI

	OLS	Within	Random
(Intercept)	52.13*** (1.78)		-8.84*** (2.50)
Population Growth	-1.79*** (0.12)	-0.14+ (0.08)	-0.20* (0.08)
Urban Population	-0.07*** (0.01)	0.34*** (0.02)	0.23*** (0.02)
Fertility Rate	-2.99*** (0.15)	-1.54*** (0.15)	-1.88*** (0.16)
ln(GDP Per Capita)	2.54*** (0.19)	8.59*** (0.29)	7.03*** (0.27)
Natural Resource Rents	-0.35*** (0.02)	0.09*** (0.02)	0.06*** (0.02)
Post-Cold War	9.87*** (0.37)	6.70*** (0.29)	7.88*** (0.28)
Num.Obs.	8252	8252	8252
R2	0.493	0.520	0.501
R2 Adj.	0.493	0.509	0.501

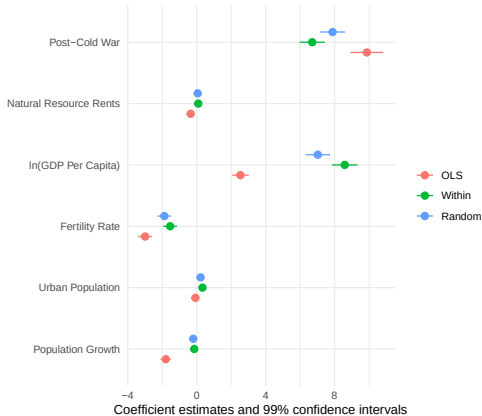
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

modelsummary: Coefficient Plots

Example code:

```
> modelplot(models, conf_level=0.99, coef_omit="(Intercept)", coef_rename=c("PopGrowth"="Population Growth",  
+ "UrbanPopulation"="Urban Population", "FertilityRate"="Fertility Rate",  
+ "log(GDPPerCapita)"="ln(GDP Per Capita)", "NaturalResourceRents"="Natural Resource Rents",  
+ "PostColdWar"="Post-Cold War"))
```

...from which we get:



Unit Effects Models: Software

R :

- the `plm` package; `plm` command
 - Fits one- and two-way FE, BE, RE models
 - Also fits first difference (FD) and instrumental variable (IV) models
- the `fixest` package; fast/scalable FE estimation for OLS and GLMs
- the `panelr` package (various commands)
- the `lme4` package; command is `lmer`
- the `nlme` package; command `lme`
- the `Paneldata` package

Stata : `xtreg`

- option `re` (the default) = random effects
- option `fe` = fixed (within) effects
- option `be` = between-effects
- Stata `package` `fect` = two-way models

Dynamics

Issues with Unit Roots in Panel Data

In general, in panel / TSCS data:

- Short series + Asymptotic tests $\rightarrow$ “borrow strength”
- Requires uniform unit roots across cross-sectional units
- Many tests require balanced panels...
- Various alternatives:
 - Maddala and Wu (1999)
 - Hadri (2000)
 - Levin, Lin and Chu (2002)
 - Im, Pesaran, and Shin (2003)
- What to do?
 - Difference the data...
 - Error-correction models

Panel Unit Root Tests for WBLI

Table: Panel Unit Root Tests: WBLI

	Test	Alternative	Statistic	Estimate	P-Value
1	Levin-Lin-Chu	stationarity	z	-1.65e16	< 0.00001
2	Hadri	at least one series has a unit root	z	199.6	< 0.00001
3	Maddala-Wu	stationarity	χ^2	294.8	0.9993
4	Im-Pesaran-Shin	stationarity	$\hat{W}_t$	-2.69e15	< 0.00001

Note: All assume individual intercepts and trends.

Consider a model like this:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\beta_{LDV} + \epsilon_{it}$$

If ϵ_{it} is perfect...

- $\hat{\beta}_{LDV}$ is biased (but consistent),
- $O(\text{bias}) = \frac{-1+3\beta_{LDV}}{T}$

If ϵ_{it} is autocorrelated...

- $\hat{\beta}_{LDV}$ is biased and inconsistent
- IV is one (bad) option...

Lagged Y s and GLS-ARMA

Can rewrite:

$$\begin{aligned}Y_{it} &= \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + u_{it} \\ u_{it} &= \phi u_{it-1} + \eta_{it}\end{aligned}$$

as

$$\begin{aligned}Y_{it} &= \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + \phi u_{it-1} + \eta_{it} \\ &= \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + \phi(Y_{it-1} - \mathbf{X}_{it-1}\boldsymbol{\beta}_{AR}) + \eta_{it} \\ &= \phi Y_{it-1} + \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + \mathbf{X}_{it-1}\psi + \eta_{it}\end{aligned}$$

where $\psi = \phi\boldsymbol{\beta}_{AR}$ and $\psi = 0$ (by assumption).

Lagged Y s and World Domination

In:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\beta_{LDV} + \epsilon_{it}$$

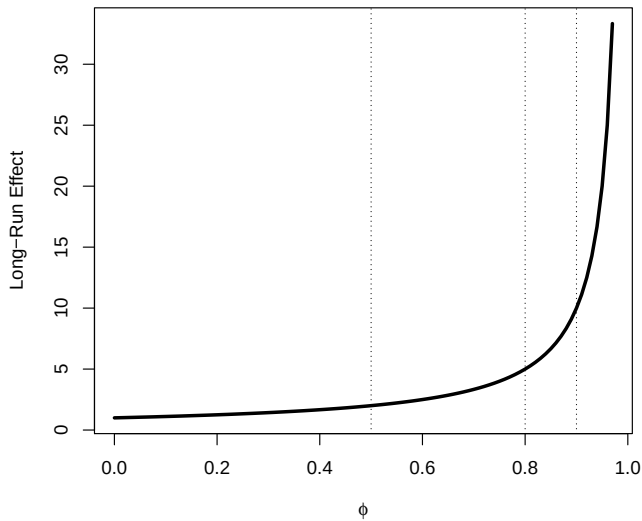
Achen: Bias “deflates” $\hat{\beta}_{LDV}$ relative to $\hat{\phi}$, “suppress” the effects of $\mathbf{X}$...

Keele & Kelly (2006):

- Contingent on ϵ s having autocorrelation
- Key: In LDV, *long-run impact of a unit change in X is:*

$$\hat{\beta}_{LR} = \frac{\hat{\beta}_{LDV}}{1 - \hat{\phi}}$$

Long-Run Impact for $\hat{\beta} = 1$



Lagged Y s and Unit Effects

Consider:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + u_{it}.$$

If we omit the unit effects, we have:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\boldsymbol{\beta} + u_{it}^*$$

with

$$u_{it}^* = \alpha_i + u_{it}$$

Lagging yields:

$$Y_{it-1} = \phi Y_{it-2} + \mathbf{X}_{it-1}\boldsymbol{\beta} + \alpha_i + u_{it-1}$$

which means

$$\text{Cov}(Y_{it-1}, u_{it}^*) \neq 0. \rightarrow \text{bias in } \hat{\phi}, \hat{\boldsymbol{\beta}}$$

Omitting fixed effects in a model with Y_{it-1} yields bias in $\hat{\phi}$ that is:

- toward zero when $\phi > 0$,
- increasing in ϕ .

Including unit effects still yields bias in $\hat{\phi}$ of $O(\frac{1}{T})$, and bias in $\hat{\beta}$.

Solutions:

- Difference/GMM estimation
- Bias correction approaches

First Difference Estimation

$$\begin{aligned}Y_{it} - Y_{it-1} &= \phi(Y_{it-1} - Y_{it-2}) + (\mathbf{X}_{it} - \mathbf{X}_{it-1})\beta + (\alpha_i - \alpha_i) + (u_{it} - u_{it-1}) \\ \Delta Y_{it} &= \phi \Delta Y_{it-1} + \Delta \mathbf{X}_{it}\beta + \Delta u_{it}\end{aligned}$$

Anderson/Hsiao: If $\nexists$ autocorrelation, then use ΔY_{it-2} or Y_{it-2} as instruments for ΔY_{it-1} ...

- Consistent in theory,
- in practice, the former is preferred, and both have issues if ϕ is high;
- both are inefficient.

Arellano & Bond (also Wawro): Use *all* lags of Y_{it} and $\mathbf{X}_{it}$ from $t - 2$ and before.

- “Good” estimates, better as $T \rightarrow \infty$,
- Easy to handle higher-order lags of Y ,
- Easy software (p1m in R , xtabond in Stata).
- Model *is* fixed effects...
- $\mathbf{Z}_i$ has $T - p - 1$ rows, $\sum_{i=p}^{T-2} i$ columns $\rightarrow$ difficulty of estimation declines in p , grows in T .

Kiviet (1995, 1999; Bun and Kiviet 2003; Bruno 2005a,b): Derive the bias in $\hat{\phi}$ and $\hat{\beta}$, then correct it...

- More accurate than the instrumental-variables/GMM estimators of A&H/A&B...
- ...especially when T is small; but not as T gets reasonably large ($T \approx 20$)

Some Dynamic Models

	Lagged Y	First Difference	FE	Lagged Y + FE
Intercept	2.204* (0.310)	0.657* (0.037)		
Lagged WBLI	0.988* (0.002)			0.956* (0.003)
Population Growth	-0.013 (0.021)	0.026 (0.025)	-0.142 (0.080)	-0.018 (0.024)
Urban Population	0.001 (0.002)	0.001 (0.051)	0.336* (0.018)	0.013* (0.006)
Fertility Rate	-0.111* (0.028)	-0.303 (0.319)	-1.543* (0.155)	-0.274* (0.047)
ln(GDP Per Capita)	-0.045 (0.033)	0.051 (0.431)	8.588* (0.289)	0.152 (0.092)
Natural Resource Rents	-0.009* (0.003)	0.019* (0.007)	0.090* (0.017)	0.001 (0.005)
Cold War	-0.215* (0.067)	0.286 (0.196)	-6.701* (0.286)	-0.294* (0.088)
R ²	0.985	0.002	0.520	0.957
Adj. R ²	0.985	0.001	0.509	0.956
Num. obs.	8140	8064	8252	8140

* $p < 0.05$

What if Y is *trending* over time?

- First Question: Why?
 - Organic growth (e.g., populations)
 - Temporary / short-term factors
 - Covariates...
- Second question: Should we care?
(A: Yes, usually... → “spurious regressions”)
- Third question: What to do?
 - Ignore it...
 - Include a counter / trend term...

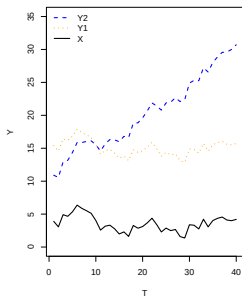
In general, adding a trend term will *decrease* the magnitudes of $\hat{\beta}$...

Trends Matter, Illustrated

Data generating processes:

$$Y_{1t} = 10 + (1 \times X_t) + u_t$$

$$Y_{2t} = 5 + (1 \times X_t) + (0.5 \times T) + u_t$$



	Dependent variable:		
	Y	Y _t	
	(1)	(2)	(3)
X	0.921*** (0.245)	-0.382 (0.786)	0.874*** (0.255)
T			0.482*** (0.026)
Constant	10.320*** (0.917)	20.220*** (2.948)	5.855*** (1.200)
Observations	40	40	40
R ²	0.272	0.006	0.905
Adjusted R ²	0.253	-0.020	0.900
Residual Std. Error	1.802 (df = 38)	5.793 (df = 38)	1.814 (df = 37)

Note:

* p<0.1; ** p<0.05; *** p<0.01

Trends Matter, Part II

Table: FE Models of WBLI

	FE	FE w/Trend	FE w/Interaction
Population Growth	-0.142* (0.080)	-0.261*** (0.067)	-0.244*** (0.067)
Urban Population	0.336*** (0.018)	0.059*** (0.016)	0.069*** (0.016)
Fertility Rate	-1.543*** (0.155)	1.659*** (0.139)	1.612*** (0.139)
ln(GDP Per Capita)	8.588*** (0.289)	2.249*** (0.263)	2.061*** (0.264)
Natural Resource Rents	0.090*** (0.017)	0.060*** (0.014)	0.059*** (0.014)
Cold War	-6.701*** (0.286)	1.943*** (0.278)	7.804*** (0.941)
Trend (1950=0)		0.739*** (0.012)	0.760*** (0.013)
Cold War x Trend			-0.171*** (0.026)
Observations	8,252	8,252	8,252
R ²	0.520	0.669	0.670
Adjusted R ²	0.509	0.661	0.662

*p<0.1; **p<0.05; ***p<0.01

R :

- the `plm` package (`purtest` for unit roots; `plm` for first-difference models)
- the `panelAR` package (GLS-ARMA models)
- the `glS` package (GLS)
- the `pgmm` package (A&B)
- the `dynpanel` package (A&H, A&B)

Stata :

- `xtgls` (GLS)
- `xtpcse` (PCSEs)
- `xtabond` / `xtdpd` (A&H A&B dynamic models)

Final Thoughts: Dynamic Panel Models

- N vs. T ...
- Are dynamics nuisance or substance?
- What problem(s) do you *really* care about?